Practical - 9

Practical Title: Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

Problem Definition: Implement secure user registration by inserting user details into MySQL using MySQLi. Validate input on frontend and backend, check for duplicate emails/usernames, hash passwords with password_hash(), and use prepared statements.

A. Theory & Analysis

1. Is password_hash() used before storing passwords?

Yes. Plaintext passwords are never saved. The registration processor uses password_hash($password, PASSWORD_DEFAULT), which generates a secure password hash using PHP's configured default algorithm (bcrypt in PHP 8.x) with an automatic salt.

2. Is duplicate email/username checked?

Yes. Before inserting a new record, the script runs a prepared SELECT statement querying the users table for matching email or username values. If a match is found, registration stops and an alert informs the user that the email is already registered.

3. Are frontend and backend validations consistent?

Yes. Both client and server check that names contain letters only (2-50 characters), email format is valid, mobile is 10 digits, and passwords have at least 6 characters with a letter and a digit. Server-side validation guarantees security even if JavaScript is disabled.

B. Implementation Details

I implemented secure user registration in php/register.php using MySQLi. The script validates input fields, queries the database for duplicate emails, hashes passwords with password_hash(), and inserts related records into both the users and students tables using an atomic transaction, while logging the action to audit_logs.

C. Evidence of Work

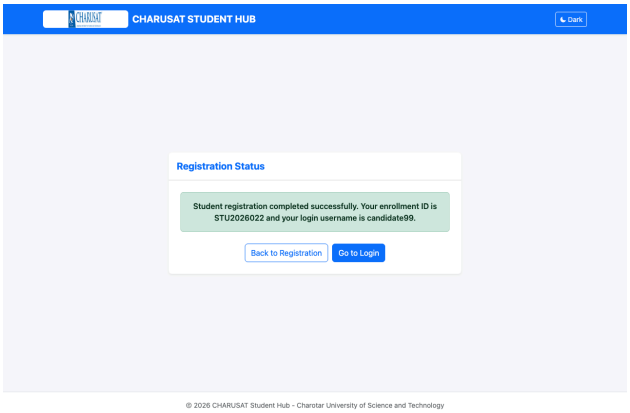


Figure 1: Successful Database Registration Confirmation with Allocated Student ID

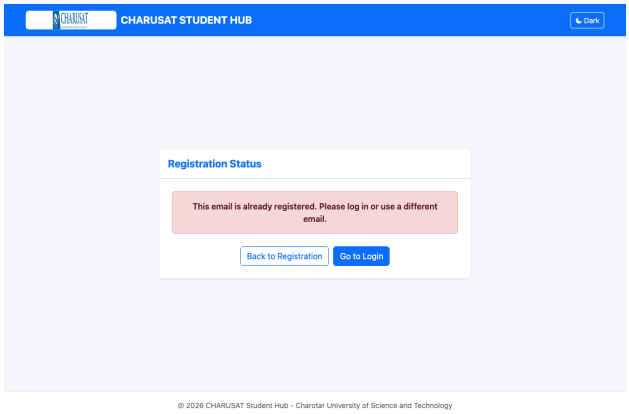


Figure 2: Duplicate Email Rejection Banner Preventing Redundant User Registration

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned how to implement secure user registration in PHP and MySQL using cryptographic password hashing, duplicate record checking, multi-table transactions, and dual-layer validation.